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DEPARTMENT OF TECHNICAL EDUCATION

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DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

COURSE PROJECT ON AUTOMATIC WATERING SYSTEM FOR PLANTS

SUBMITTED BY

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ABSTRACT

Watering is the most important cultural practice and most labor intensive task in daily greenhouse operation. Watering systems ease the burden of getting water to plants when they need it. Knowing when and how much to water is two important aspects of watering process. To make the gardener works easily, the automatic plant watering system is created. There have a various type using automatic watering system that are by using sprinkler system, tube, nozzles and other. This project uses watering sprinkler system because it can water the plants located in the pots. This project uses Arduino board, which consists of ATmega328 Microcontroller. It is programmed in such a way that it will sense the moisture level of the plants and supply the water if required. This type of system is often used for general plant care, as part of caring for small and large gardens. Normally, the plants need to be watered twice daily, morning and evening. So, the microcontroller has to be coded to water the plants in the greenhouse about two times per day. People enjoy plants, their benefits and the feeling related to nurturing them. However for most people it becomes challenging to keep them healthy and alive. To accommodate this challenge we have developed a prototype, which makes a plant more self-sufficient, watering itself from a large water tank and providing itself with artificial sunlight. The pro-To type reports status of its current conditions and also reminds the user to refill the water tank. The system www.jespublication.com automation is designed to be assistive to the user. We hope that through this prototype people will enjoy having plants without the challenges related to absent or forgetfulness.

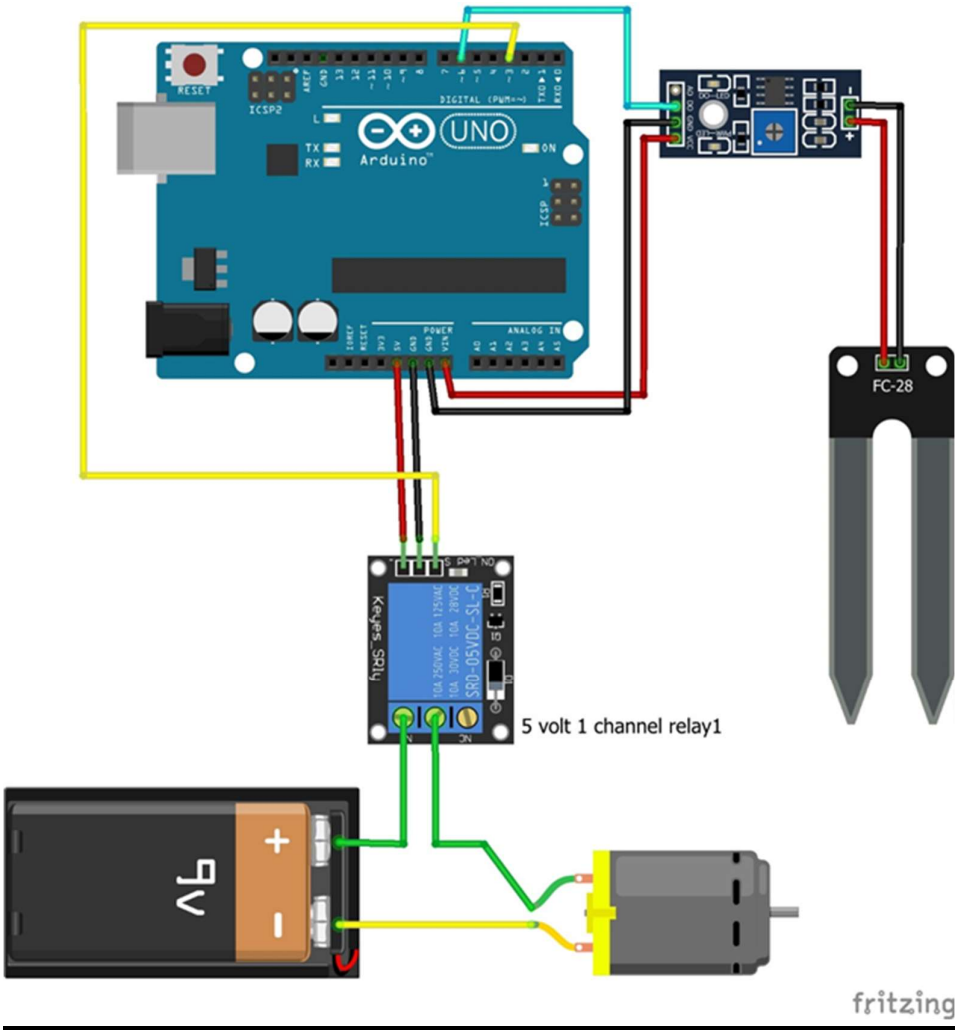
INTRODUCTION

The project aims in designing a system which is capable of detecting moisture level in the soil and capable of taking the decision of switching ON/ OFF water motor. The Arduino Uno is a microcontroller board based on the ATmega328. It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz ceramic resonator, a USB connection, a power jack, an ICSP header, and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with a AC-to-DC adapter or battery to get started. The system makes use of a microcontroller. The output from soil moisture sensor is fed to microcontroller will continuously compare output from soil moisture sensor when there is no moisture in soil. The microcontroller gets on the water motor through driver.

OVER VIEW:-

An embedded system is a combination of software and hardware to perform a dedicated task. Some of the main devices used in embedded products are Microprocessors and Microcontrollers. Microprocessors are commonly referred to as general purpose processors as they simply accept the inputs, process it and give the output. In contrast, a microcontroller not only accepts the data as inputs but also manipulates it, interfaces the data with various devices, controls the data and thus finally gives the result. The “Arduino based automatic plant watering system” using arduino microcontroller is an exclusive project which is used to switch on/off the water pump automatically when the soil moisture sensor detects to the plant. We can use this project to reduce deaths of people due to current shocks at fields and also to on/off the motor automatically by using soil moisture sensors at fields which avoids need of a human being .

CIRCUIT DIAGRAM

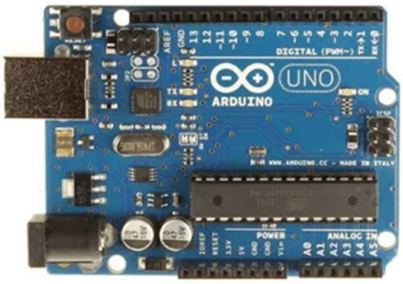


COMPONENTS

The components we used for this project are:

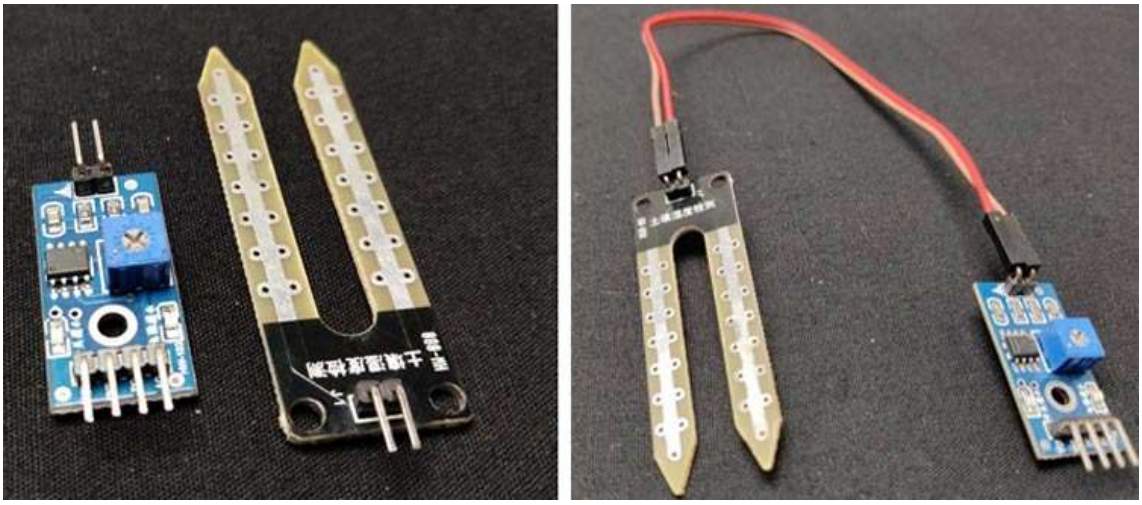
1. Arduino UNO
2. Soil sensor
3. 9V Battery
4. 9V Mini water pump
5. 5V relay module.

ARDUINO UNO:-



Arduino UNO is a microcontroller board based on the **ATmega328P**. It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz ceramic resonator, a USB connection, a power jack, an ICSP header and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with a AC-to-DC adapter or battery to get started. You can tinker with your UNO without worrying too much about doing something wrong, worst case scenario you can replace the chip for a few dollars and start over again.

SOIL SENSOR:-



The working of the soil moisture sensor is very easy to understand. It has 2 probes with exposed contacts that act like a **variable resistor** whose **resistance varies** according to the water content in the soil. This resistance is inversely proportional to the soil moisture which means that higher water in the soil means better conductivity and hence a lower resistance. While the lower water in the soil means poor conductivity and will result in higher resistance. The sensor produces an analog voltage output according to the resistance.

The sensor comes with an electronic module that connects the probe to the Arduino. The module has an **LM393 High Precision Comparator** which converts the analog signal to a Digital Output which is fed to the microcontroller. We have covered an in-depth Arduino soil moisture sensor tutorial which covers the working of soil moisture sensor module and how to use it with the Arduino. You can check the tutorial if you want to learn more about the soil moisture sensor.

DC WATER PUMP :-



We need a small pump to irrigate the plant, but in the case of a garden, we need to drive a larger pump that can provide a higher volume of water depending on the size of your garden which can't be directly powered by an Arduino. So in case you need to operate a larger pump, a driver is necessary to provide enough current for the pump, to show that I am using a 5v relay. You can also use an AC-powered pump and use a suitable relay. The working will remain the same as shown in this project, you just have to replace the DC power input connected to the relay with an AC power input and have to power your Arduino with a separate DC power source.

WORKING

Sensing Soil Moisture:-

The soil moisture sensor is placed in the soil near the plants. This sensor continuously checks the soil's moisture level and sends the data to the microcontroller.

The microcontroller has a threshold set for soil dryness, which determines when the plants need watering.

Activating the Pump:-

When the soil is dry (below the set moisture threshold), the microcontroller activates the water pump.

Water is pumped from the reservoir, through the tubing, and into the soil near the plants.

Water Delivery:-

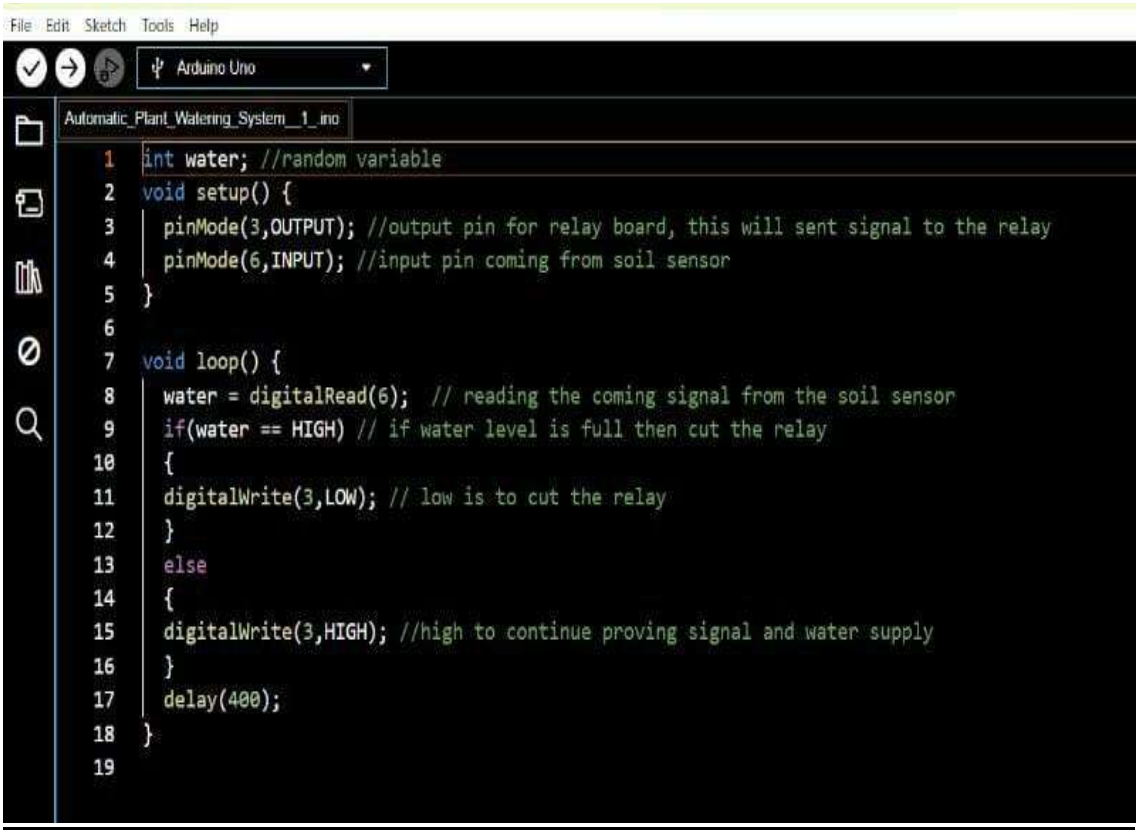
Water flows through the tubes and is distributed evenly around the plants using drippers. The amount of water can be adjusted based on the needs of different plants.

The pump continues to operate until the moisture sensor detects sufficient moisture in the soil.

Shutting Off the System:-

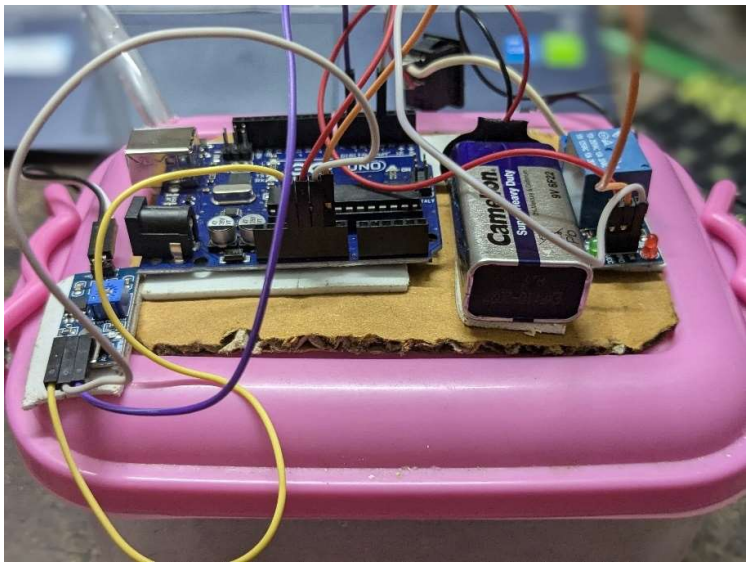
When the soil reaches the desired moisture level, the sensor detects this change, and the microcontroller sends a signal to turn off the pump.

This prevents overwatering and conserves water, making the system efficient and eco-friendly.


File Edit Sketch Tools Help
Arduino Uno
Automatic_Plant_Watering_System__1_.ino
int water; //random variable
void setup() {
pinMode(3,OUTPUT); //output pin for relay board, this will sent signal to the relay
pinMode(6,INPUT); //input pin coming from soil sensor
}
void loop() {
water = digitalRead(6); // reading the coming signal from the soil sensor
if(water == HIGH) // if water level is full then cut the relay
{
digitalWrite(3,LOW); // low is to cut the relay
}
else
{
digitalWrite(3,HIGH); //high to continue proving signal and water supply
}
delay(400);
}

RESULT

The project “Arduino based automatic plant watering system” is designed to operate a water pump automatically based on the soil moisture sensor detection of sufficient water to the plant or in fields. This project eliminates the manual switching mechanism used by the farmers to ON/OFF the irrigation system.



CONCLUSION

Integrating features of all the hardware components used have been developed in it. Presence of every module has been reasoned out and placed carefully, thus contributing to the best working of the unit. Secondly, using highly advanced IC’s with the help of growing technology, the project has been successfully implemented. Thus the project has been successfully designed and tested.

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